

Diego Becerra Fuentes

Mechatronics Engineering Student

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Education

09/2026 - Present **Erasmus Double-Degree Program**, *Hochschule Esslingen*, Esslingen am Neckar, Germany
- Mechanical Engineering, Automation Technology and Manufacturing Informatics

08/2023 - Present **Mechatronics Engineering**, *Tecnológico de Monterrey*, Monterrey, Mexico
- GPA: 90/100
- Relevant courses: Calculus, Statistics, Mechanics, Manufacturing Methods, Dynamic Systems, Electronics, Industrial Automation, PLCs, Control Systems Design, Robotic Design, Industrial Communication

Available from 03/2027 for a 6 month mandatory internship including bachelor's thesis

Engineering Projects

09/2026 - Present **Battle Bot Design and Construction**, *Esslingen University of Applied Sciences*
- Design and construction of a 1.5 kg Battle Bot to compete against other teams at Esslingen University, due 01/2027.
- Design with constraints and within a budget; industrial laser cutting, machining and metal bending parts will be manufactured with university equipment.

04/2026 **Digital Synthesizer**, *Personal Project*
- STM32 powered digital synth using both resistive and capacitive touch sensors as input devices.
- Custom PCBs were designed for the project.
- Design and construction of the synthesizer body, made with laser-cut MDF and a 3D printed structural interior.
- Resistive touch sensor made with laser-cut film of conductive plastic.
- Programming in C of digital-to-analog converters to generate digital oscillators and filters.

12/2025 **Autonomous Scaled Tractor**, *John Deere*, Training Partner
- Design, calibration, construction, and testing of an STM32 Nucleo-F103RB powered scale autonomous vehicle, using an IMU to detect heading and an ultrasonic sensor to detect and avoid obstacles.
- Programmed microcontroller in C using an RTOS to manage sensor inputs and calculate heading with a Fuzzy Logic Controller and PID steering control.

10/2025 **Analog Temperature Controller**, *John Deere*, Training Partner
- Designed, built and simulated a chamber representing the insides of a John Deere tractor to determine how it dissipates heat.
- Simulated heat and air flow using SolidWorks Thermal analysis, taking into account heat sources and internal ventilation.
- Design and construction of an analog temperature controller using op amps to create a Schmitt trigger that regulates temperature inside of the built chamber.

10/2025 **Self-Balancing Wing**, *University Project*
- Design, construction, and testing of an ESP32 powered self-balancing wing with an IMU as input and a servomotor controlling the wing's ailerons.
- The wing and fuselage were built of a combination of MDF and 3D printed parts, and the wings had a paper skin.
- The PID parameters were calculated using MATLAB Simulink. The physical device was too large to test in campus wind tunnels but showed critical stability when put in front of a running fan.

Competitions and Awards

06/2026 **1st Place out of 7 Teams – Digital Twin of an Electrical Components Production Line**, *Schneider Electric*, Training Partner

- Design, programming and construction of a Collaborative Robot powered mock industrial cell with industrial imaging and piece selection/rejection capabilities built using ROS, OpenCV, and TIA Portal.
- Design and programming of a digital twin of said industrial cell with operation simulation, remote telemetry and remote operation capability using a Raspberry Pi 4.

Student Communities and Community Service

2 semesters **SAERO Aeronautical Society** – Research Subdirector

- Ion propulsion RC aircraft project: Worked constructing the airplane´s body and with electronic component procurement.
- Compliant airfoil RC aircraft project: Worked on design, construction and mechanical testing of compliant airfoils.

2 semesters **Mty Rocket Lab** – Avionics Area

- Worked on electronic module procurement, electronic troubleshooting and soldering.

Game Jams

- Participated in several game jams.

10 weeks **Sosac Foundation**

- Surveyed illegal dumpsites and took part in river cleanup.

10 weeks **F21 Foundation**

- Distributed educational supplies to underprivileged communities.

15 weeks **Jann Foundation**

- Tutored mathematics online for underserved middle-school students.

Certifications

2026 CSWA – Certified SolidWorks Associate

2026 CSWP – Certified SolidWorks Professional

Skills

Languages

Spanish	Native (C2)	
English	Fluent (C1)	
German	Basic (A2)	*B1 certificate expected by 01/2027*

Technical

Programming	C/C++, Python, MATLAB
Embedded	STM32, ESP32, Arduino
Robotics / PLC	PLCs, Siemens TIA Portal, ABB RobotStudio, PID control, industrial communication
Machine Vision	OpenCV
CAD / Mfg.	SolidWorks, design for manufacturing, 3D printing, laser cutting, CNC machining, welding
Electronics	PCB design (KiCad, Proteus), circuit assembly and soldering
Machine Vision	OpenCV, Cognex In-Sight Explorer
Data Analysis	Minitab, RStudio

Esslingen am Neckar, 31.10.2026, Diego Becerra Fuentes